

# AI/ML intern DAY 4

## Task 3 — Body Segmentation Methods Research

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### 1. Objective

The objective of this task was to research and understand body segmentation methods used in Artificial Intelligence and computer vision systems.

The research focused on:

- human body segmentation
- semantic segmentation
- instance segmentation
- segmentation applications in fashion technology
- AI-based body analysis systems

The task also explored how segmentation improves virtual try-on systems and body tracking applications.

### 2. Introduction to Body Segmentation

Body segmentation is a computer vision technique used to separate the human body from the background in images or videos.

Segmentation helps AI systems:

- identify body shape
- track body movement
- analyze posture
- improve clothing fitting accuracy

Body segmentation plays an important role in:

- virtual try-on systems
- pose estimation
- healthcare systems
- fitness tracking
- AI fashion applications

The segmentation process helps machines understand different body regions and improves real-time body analysis.

### **3. Types of Body Segmentation**

#### **1. Semantic Segmentation**

Semantic segmentation classifies every pixel in an image into categories such as:

- body
- clothing
- background

The AI system identifies which pixels belong to the human body and separates them from the environment.

#### **Applications**

- background removal
- virtual fitting rooms
- image editing
- body tracking systems

#### **2. Instance Segmentation**

Instance segmentation identifies separate objects individually.

Example:

- detecting multiple people separately in one image

This method helps AI systems track different individuals independently.

#### **Applications**

- crowd analysis
- multi-person tracking
- surveillance systems
- advanced AI vision systems

### 3. Human Body Segmentation

Human body segmentation focuses specifically on detecting:

- torso
- arms
- legs
- face
- body boundaries

This improves:

- clothing alignment
- body analysis
- pose estimation accuracy



## **4. Working of Body Segmentation Systems**

Body segmentation systems work using the following steps:

### **Step 1 — Image Capture**

The AI system captures images or live video frames using a webcam or camera.

### **Step 2 — Image Processing**

The image is processed using computer vision algorithms and deep learning models.

### **Step 3 — Pixel Classification**

The AI model classifies pixels into:

- body
- clothing
- background

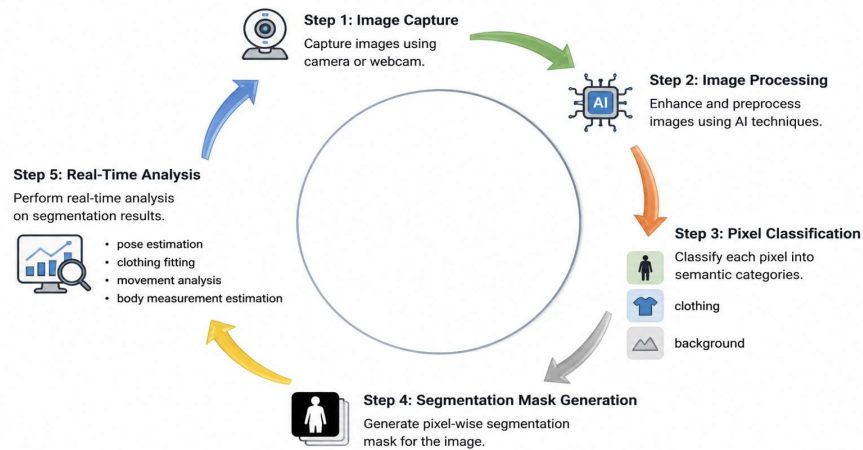
### **Step 4 — Segmentation Mask Generation**

A segmentation mask is created to separate the human body from the environment.

### **Step 5 — Real-Time Analysis**

The segmented body is used for:

- pose estimation
- clothing fitting
- movement analysis
- body measurement estimation



## 5. Applications of Body Segmentation

Body segmentation systems are used in:

- virtual fitting rooms
- AI fashion systems
- healthcare applications
- fitness tracking
- background removal
- motion tracking
- body measurement estimation
- augmented reality systems

### Fashion Technology

Segmentation improves:

- clothing overlay accuracy
- body measurement estimation
- virtual try-on systems

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### Healthcare Applications

Segmentation helps monitor:

- posture

- rehabilitation exercises
- body movement analysis

## 6. Advantages of Body Segmentation

Body segmentation provides several advantages:

- accurate body analysis
- improved clothing fitting
- better AI tracking systems
- enhanced pose estimation
- real-time body detection
- better user interaction
- improved computer vision performance

## 7. Challenges in Body Segmentation

| Challenge             | Description                                                                  | Practical Mitigation                                                                                                                                                          |
|-----------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lighting variations   | Changes in brightness/colour cast reduce segmentation and landmark quality.  | Guide user to uniform, front-lit environment; apply histogram equalization / CLAHE and white-balance preprocessing; use temporal smoothing to recover brief drops in quality. |
| Background complexity | Busy or similar-toned backgrounds make separating body from scene difficult. | Ask for a plain background or use background subtraction (running median) and person segmentation models (semantic mask) before pose estimation.                              |
| Real-time processing  | High-resolution segmentation/models cause latency and dropped frames.        | Resize frames, run lightweight or quantized models, lower inference rate (e.g., process every Nth frame), and use asynchronous pipelines or GPU acceleration.                 |

|                     |                                                                                |                                                                                                                                                                                                         |
|---------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Occlusion problems  | Body parts hidden (self-occlusion or by objects) break landmark continuity.    | Use temporal interpolation/smoothing, pose priors (kinematic constraints), request re-poses, or fuse multiple views if available.                                                                       |
| Clothing similarity | Clothing colours/patterns close to skin or background obscure body boundaries. | Encourage form-fitting or contrasting clothing during capture, detect and warn about loose/ambiguous garments, and consider clothing-aware segmentation models or multimodal cues (depth if available). |

## 8. Learning Outcomes

During this research task, the following concepts were learned:

- semantic segmentation
- instance segmentation
- body mask generation
- AI-based body analysis
- segmentation workflows
- fashion AI applications
- computer vision segmentation systems
- real-time AI processing

## 9. Conclusion

Body segmentation is an important computer vision technology that helps AI systems separate human body regions from the background for accurate body analysis.

The research demonstrated how segmentation improves:

- virtual try-on systems
- body tracking
- pose estimation

- AI fashion technologies

The task also improved understanding of:

- segmentation methods
- AI workflows
- deep learning applications
- real-time body analysis systems

Body segmentation plays a major role in modern AI-powered computer vision applications.

